Requirement Analysis

Solution Requirements

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Input Collection	Accept 7 numeric inputs from the user: N, P, K, temperature, humidity, pH, rainfall via a web form
FR-2	Input Validation	Validate that all entered values are within realistic agricultural ranges
FR-3	Crop Prediction	Use the trained ML model to predict the most suitable crop for the given inputs
FR-4	Result Display	Display the predicted crop name and confidence score clearly to the user
FR-5	Alternate Suggestions	Show the top-3 alternate crop suggestions with their confidence percentages
FR-6	Error Handling	Show a clear, readable error message for invalid or missing inputs instead of crashing

Non-Functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple form usable by non-technical users (farmers) without any training
NFR-2	Performance	Returns a crop prediction result within 2-3 seconds of submission
NFR-3	Reliability	Model achieves at least 90% prediction accuracy on test data

NFR-4	Availability	Accessible anytime via a public deployed link, no installation required
NFR-5	Scalability	Can be extended with more crops, fertilizer tips, and regional language support in future